GEODE Datathon

Team: Operation Hotspot

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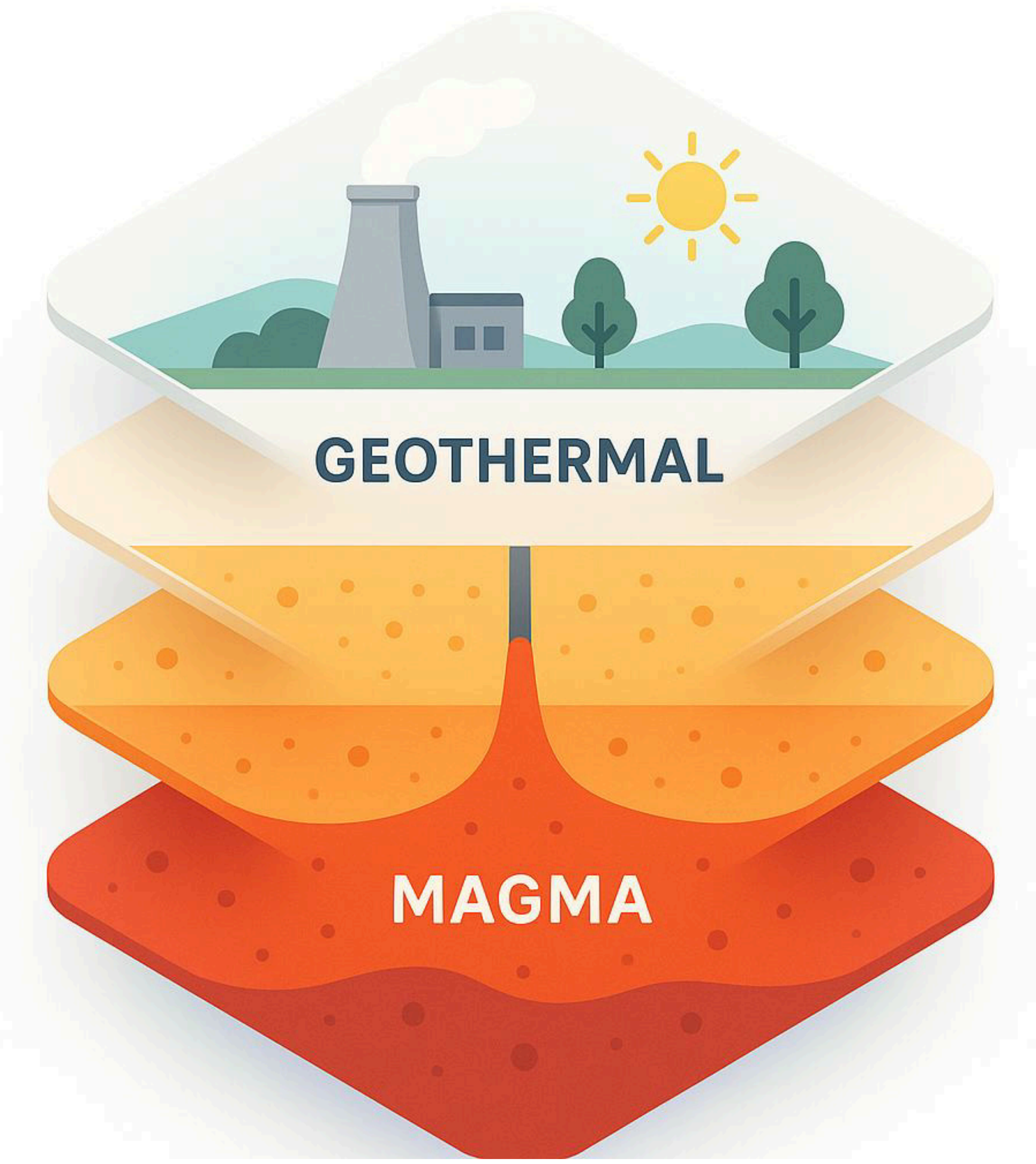


Geoscience Challenge

Agenda

The Challenge

Nevada PFA Project: Identification of blind geothermal systems in Nevada's Great Basin region in areas lacking surface geothermal indicators

Challenge: Can we leverage advanced AI to refine/ augment these complex spatial analyses?

Our Goal: •Develop a data-driven spatial learning approach to **learn from the PFA, validate** its findings, and potentially **identify** new prospective zones.

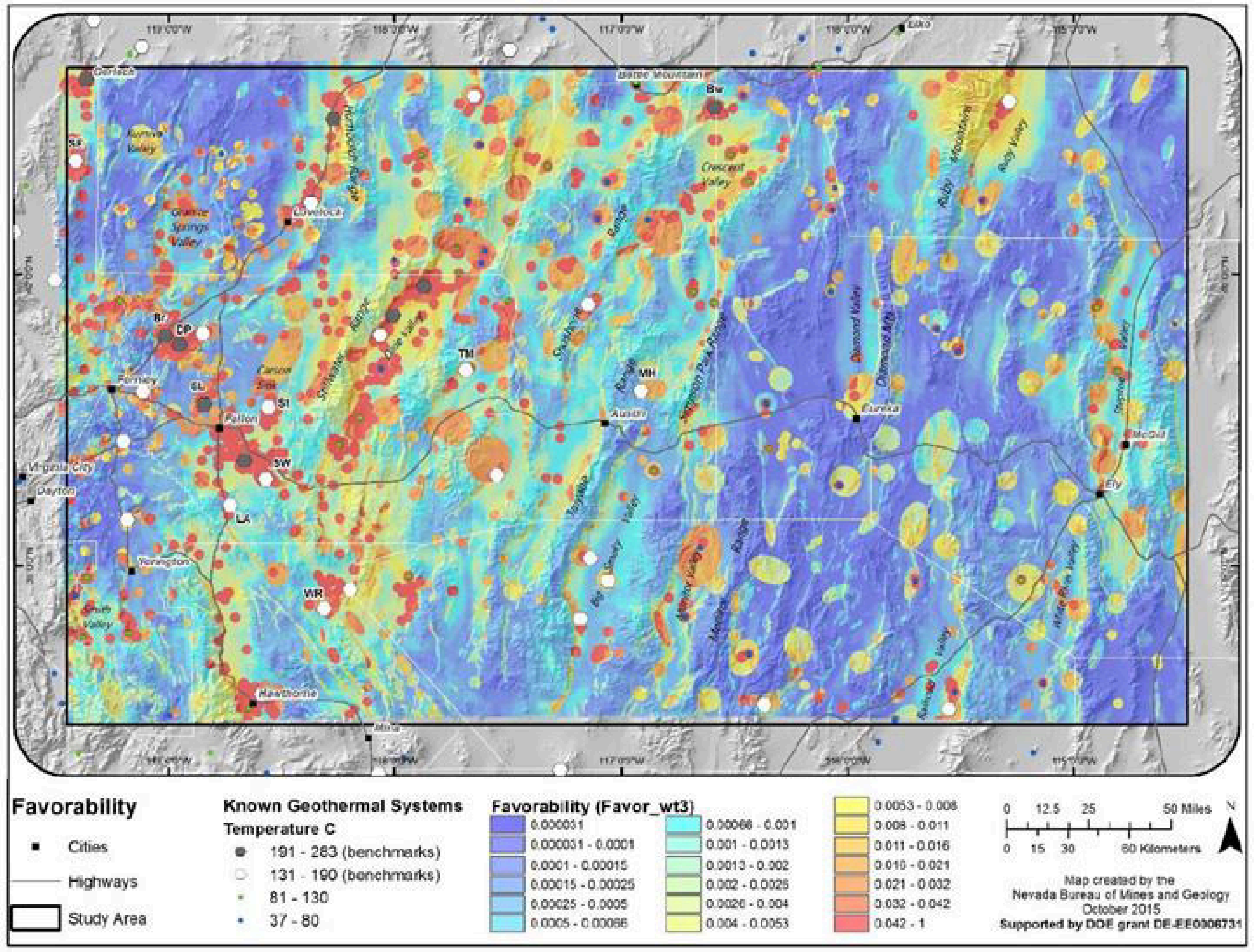


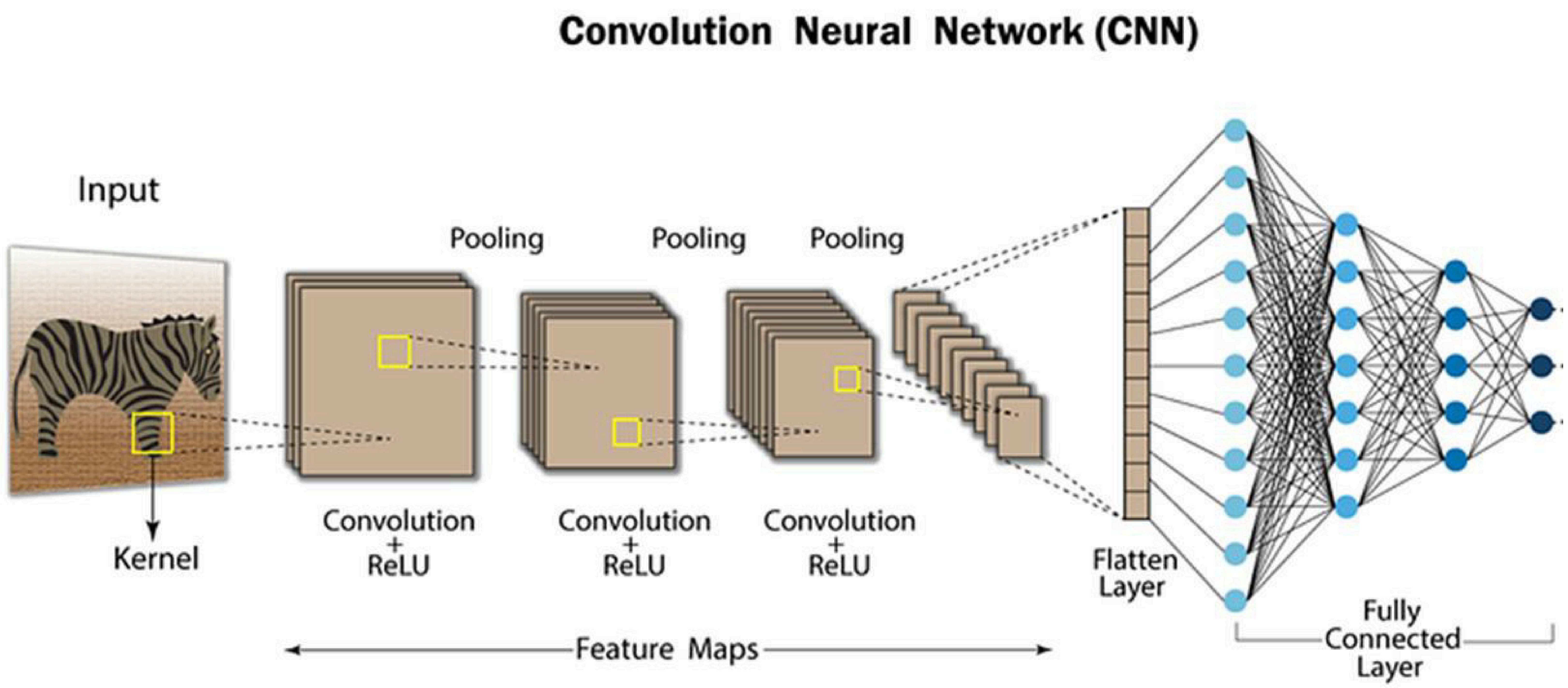
Figure 34. The favorability model generated by combination of the play fairway model (Fig. M-29) with the direct evidence model (Fig. M-32).

Overall favorability map of the Nevada PFA study. Retrieved from Faulds et al (2015)

Our Approach

Convolutional Neural Networks

- We wanted to put something over the existing research.
- **CNN is the next step (Smith et al, 2023).**
- **Leverage** the **existing PFA model** to build new research avenues.

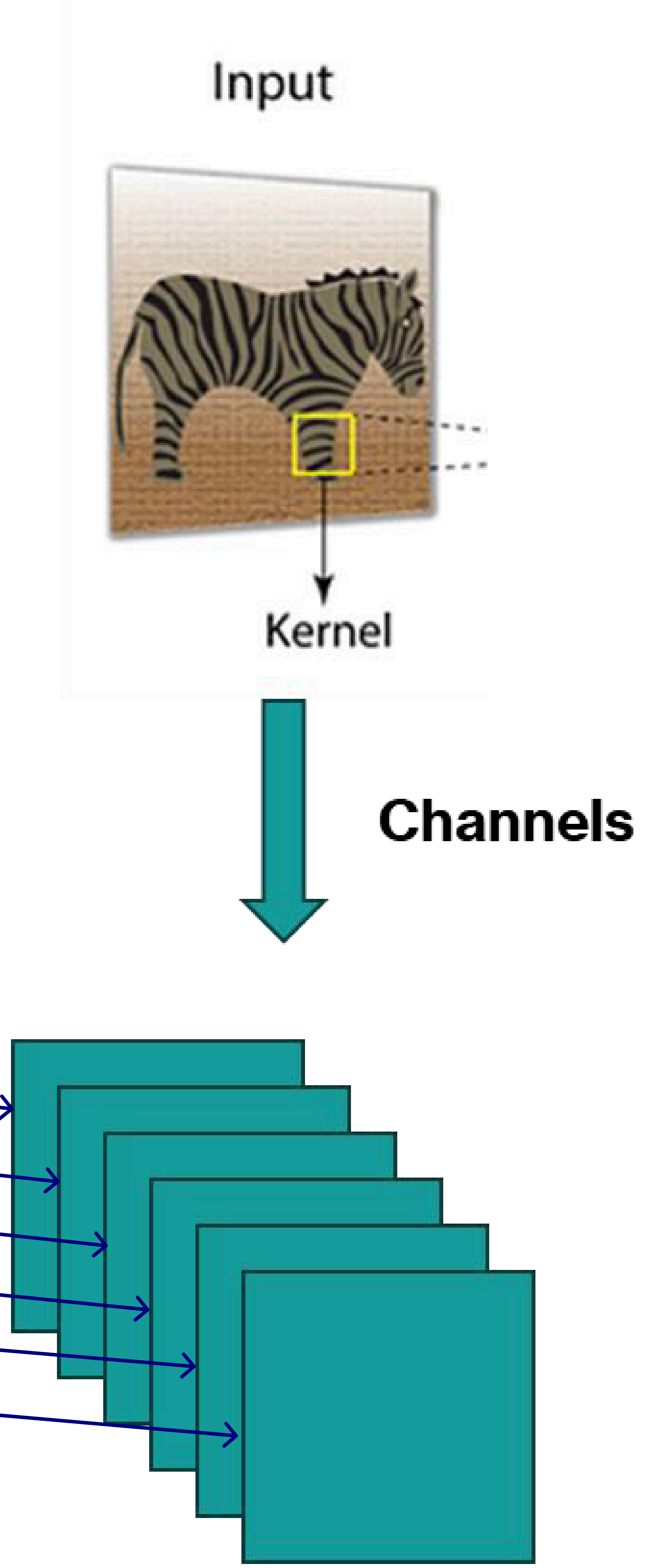


A typical CNN architecture. Retrieved from https://nafizshahriar.medium.com

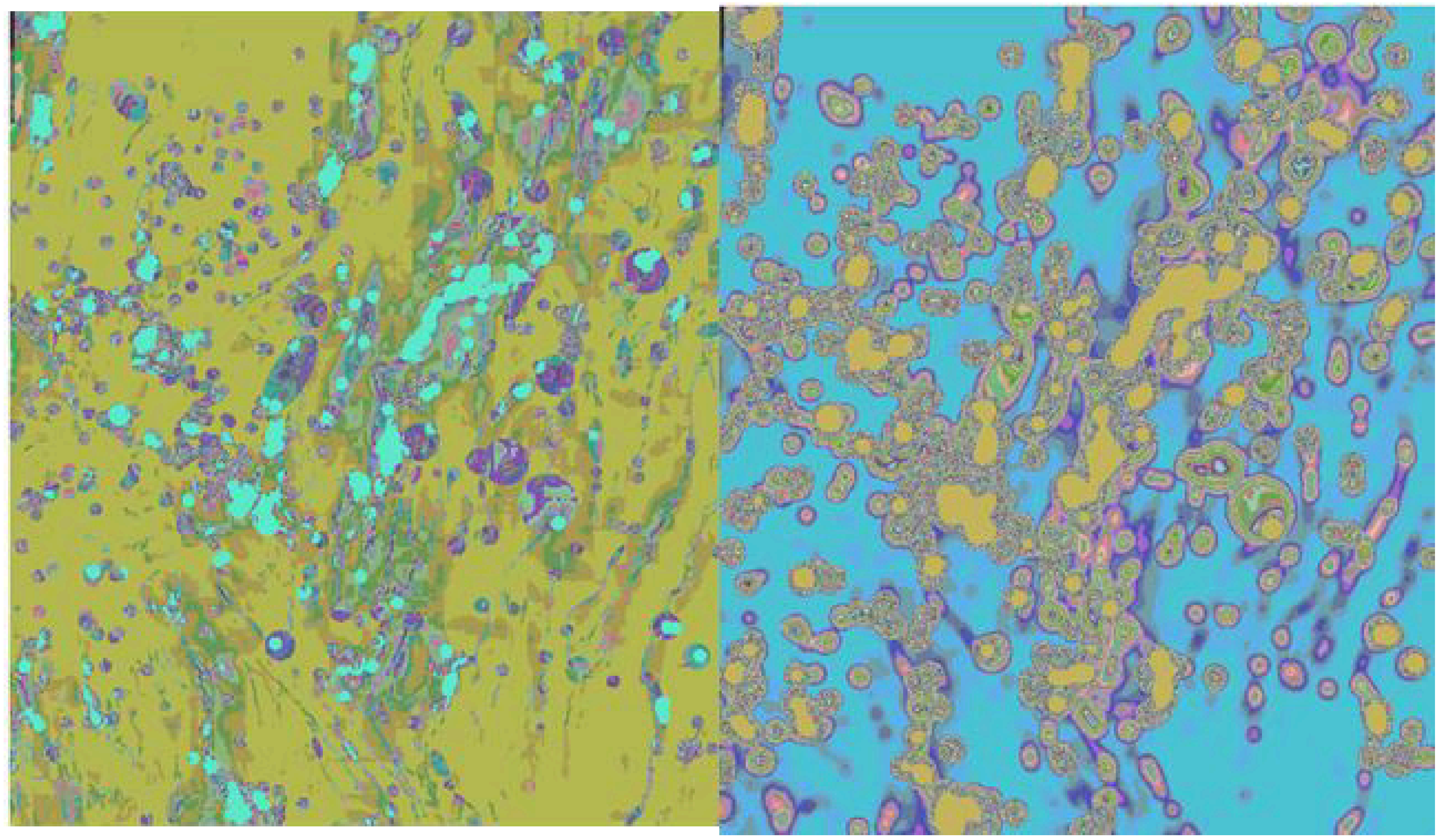
Feature Engineering

Response Feature
•Overall Favorability – **hitchhiking** with the existing work.
•Model can learn underlying spatial patterns that leads to high favorability.
•Opportunity to validate & identify.

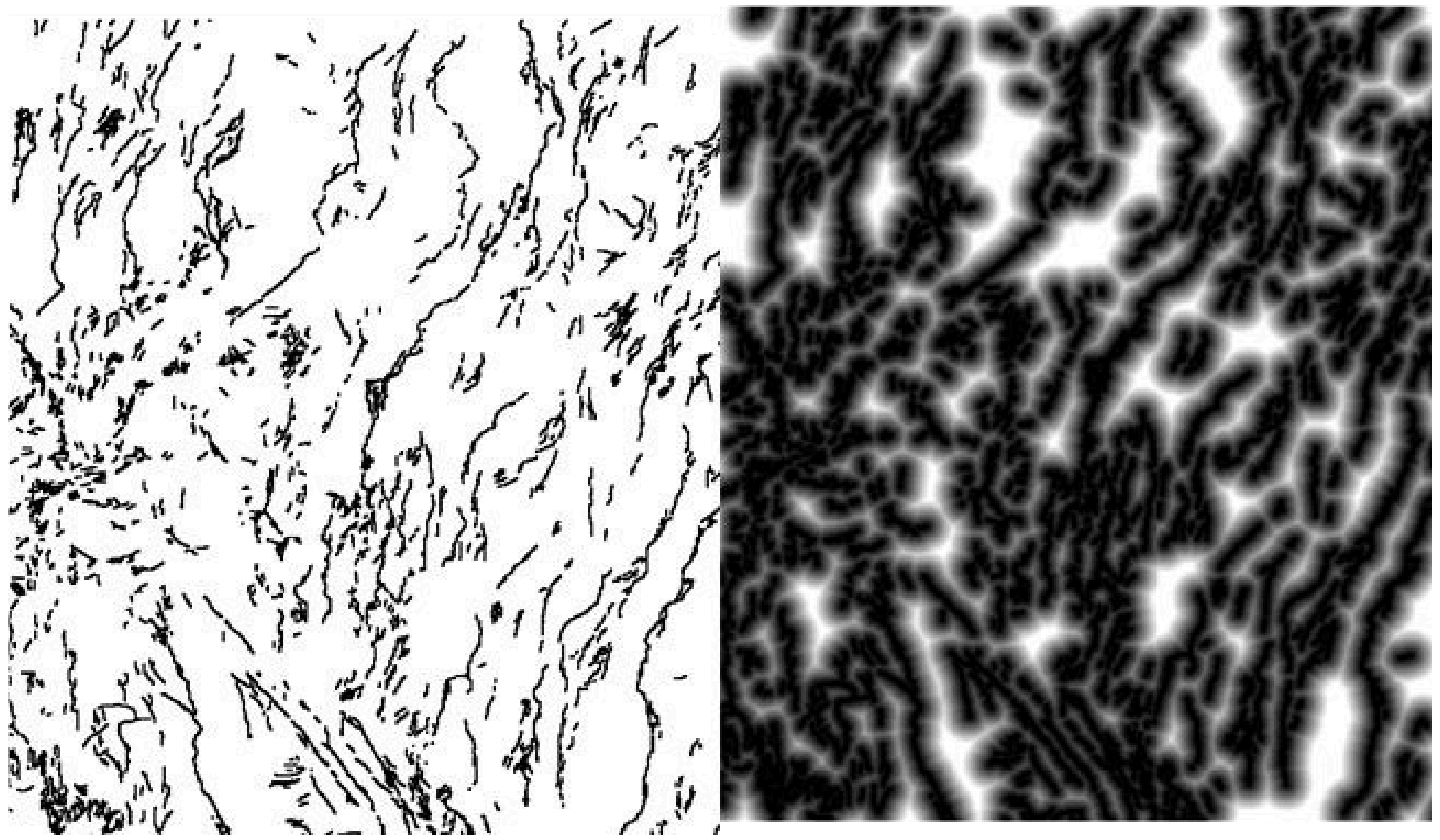
Input Features
•Geodetic strain
•Horizontal gravity
•Regional permeability
•Temperature at 3km
•Earthquake density
•Faults (500m Buffer)



Data Preprocessing



Raw Favorability (left) and Smoothed Favorability (right)



Raw Faults (left) and Euclidean Distance to Fault at Location (right)

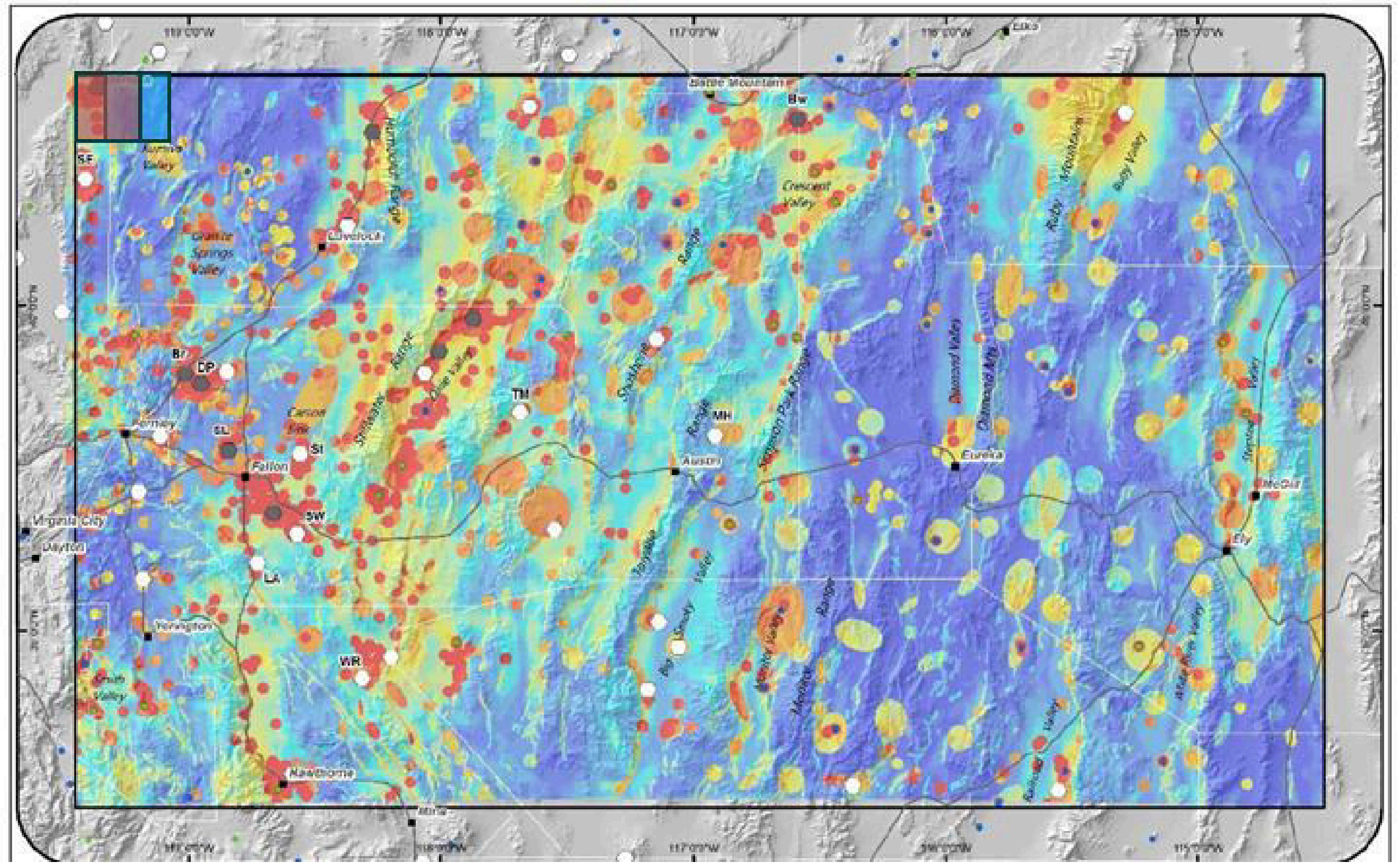
Overall Favorability

- **Biases** because of the human-identified areas – sharp transitions.
- **Focal Statistics** to blur.

Faults

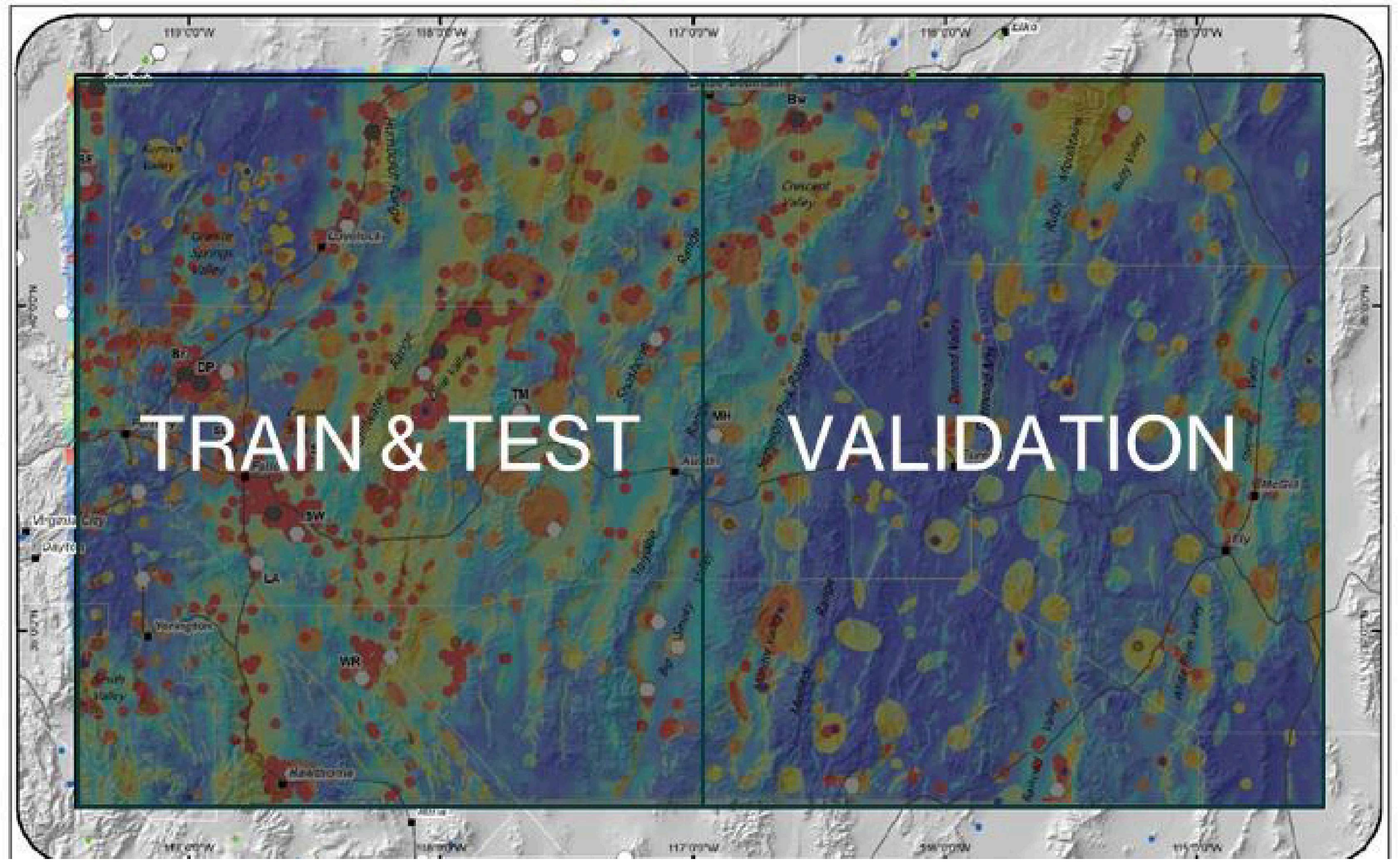
- Too sparse if used as-is.
- **Euclidean distance** to find a value for the entire map.

Methodology



CNN Overview

- **Pixel-to-pixel** regression CNN.
- 32×32 patches
- Sliding 2 pixels right & down for sampling



Splitting of the data into train, test and validation sets

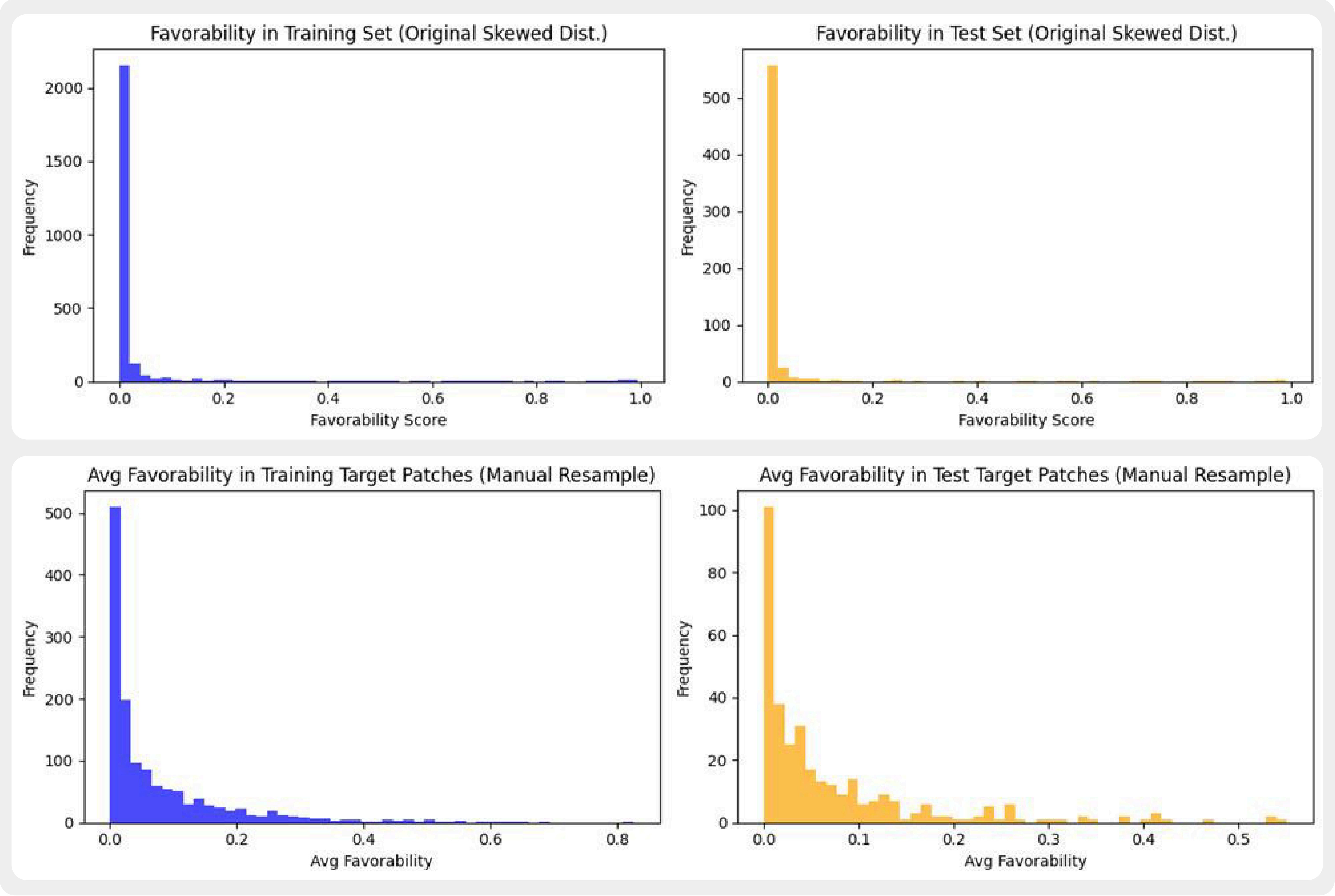
Evaluation Strategy

- **Left Half** used as **train-test**
- **Right Half** used as **validation**
- Pixel-to-pixel MSE
  - Quantitative results are NOT aimed – we are looking for relatively high favorability areas.

Methodology – Resampling

Resampling from the high favorability areas

Defined manual bins of favorability score. Picked **more samples** from the **high favorability** score bins.



Favorability distribution of the training and testing samples before resampling (Up) and after resampling (high)

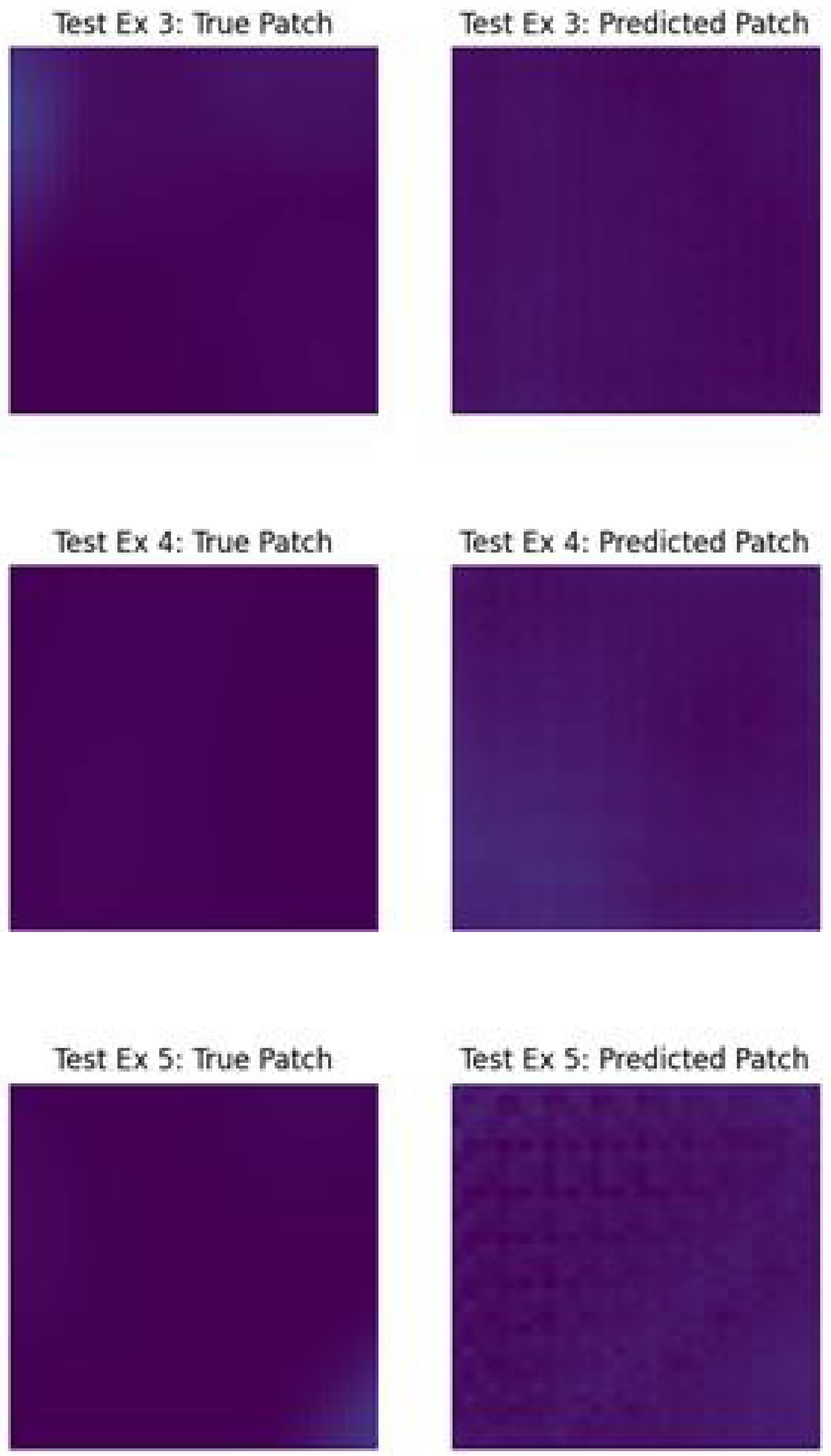
Results – Train & Test

Model learns the relationship in ~50 epochs

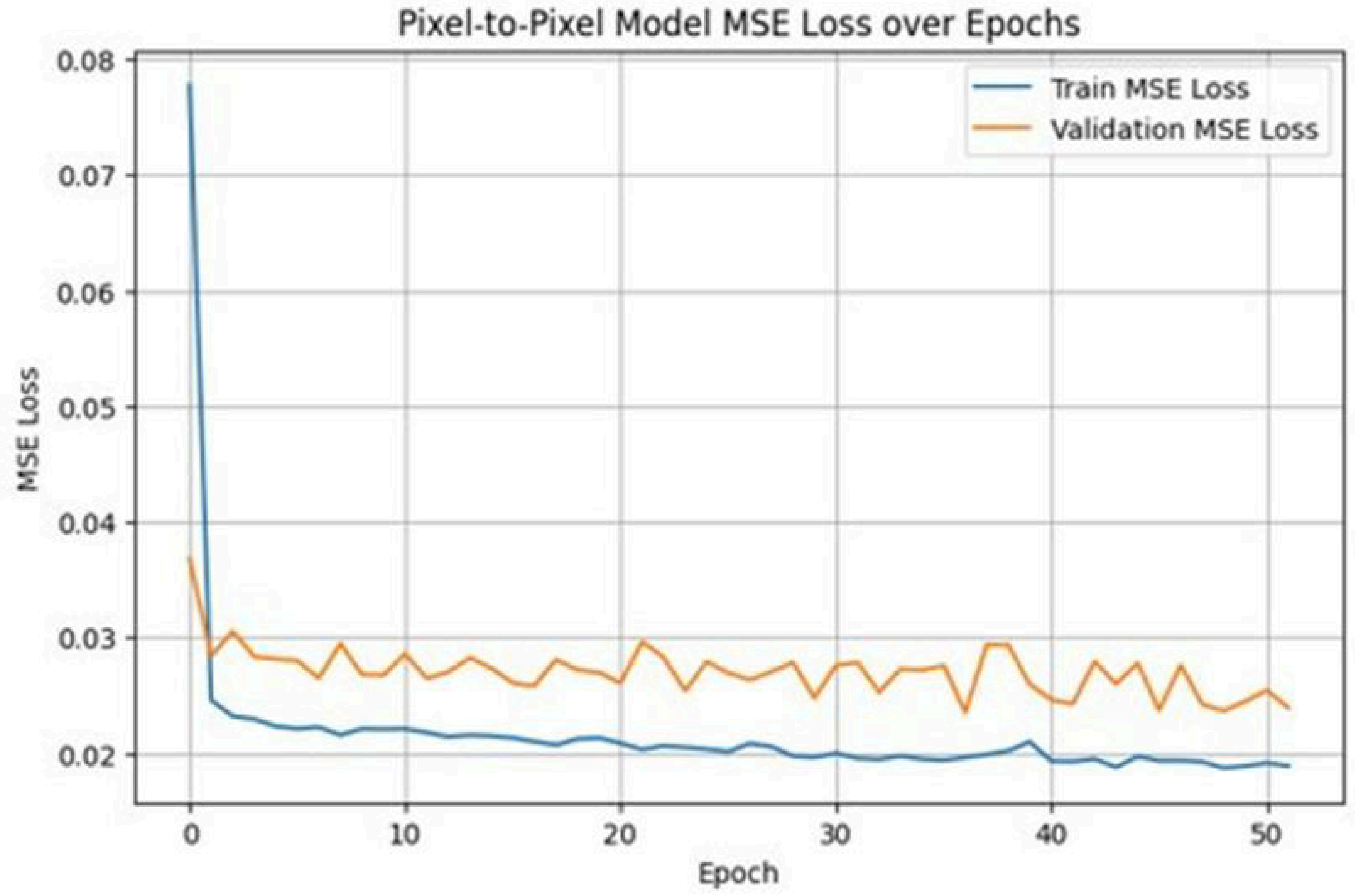
- MSE values per se don't mean anything – we need to see the visual results.

The model can capture the relationship locally

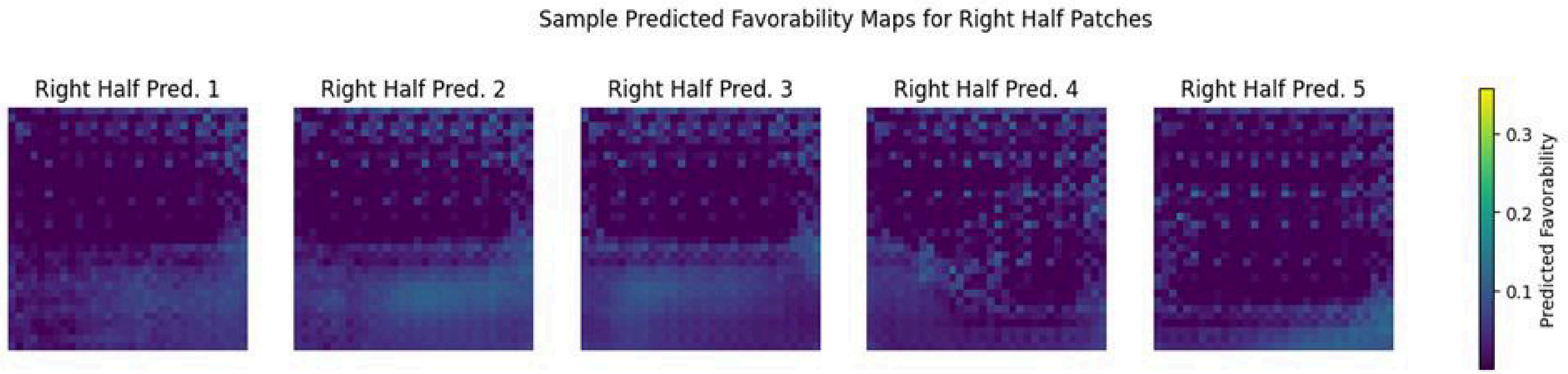
- But what about on the 'global' scale?



A few random test patches. True (left) and predicted (right)



MSE loss over epochs

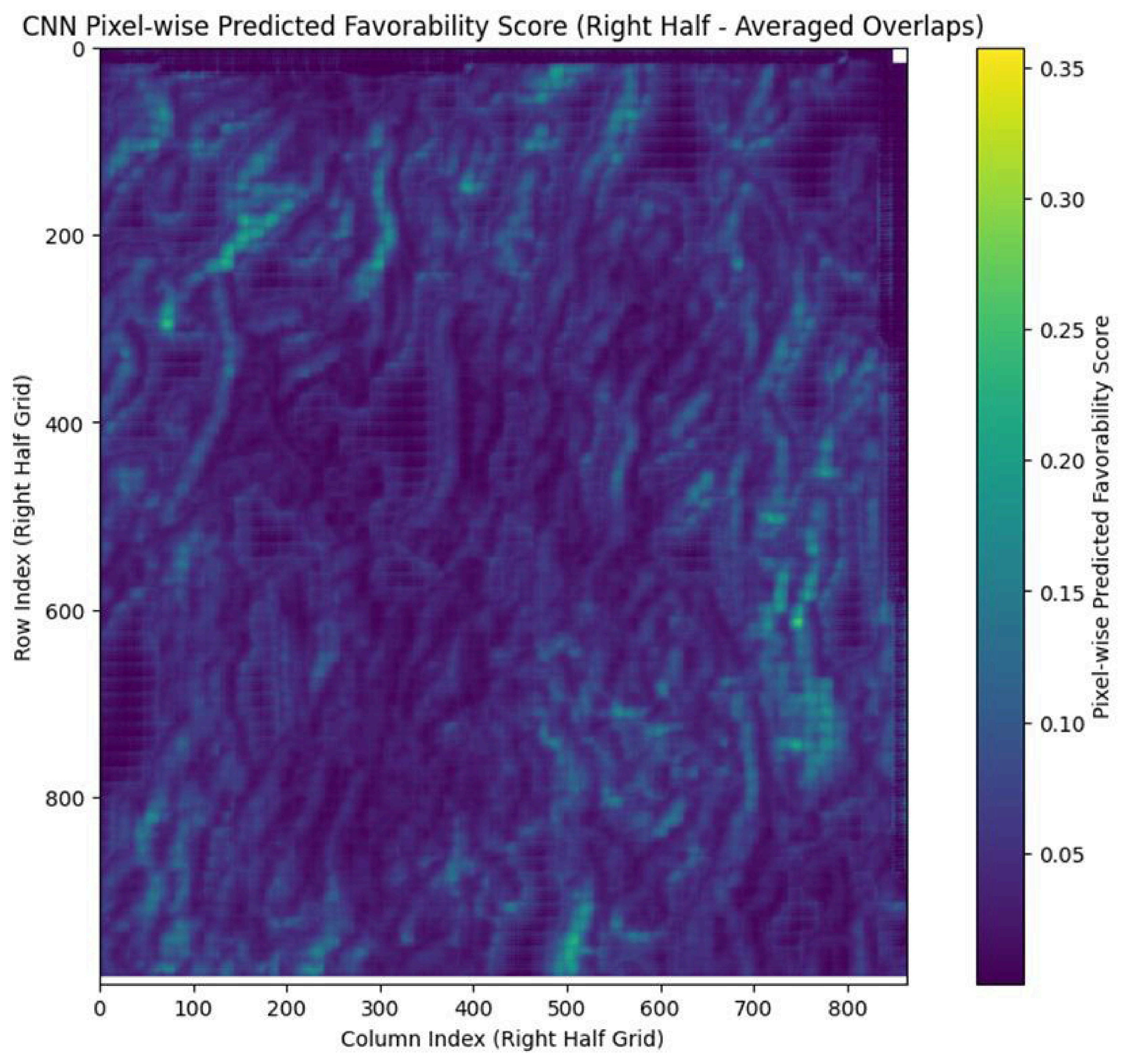


A few random validation (right half of the map) patches

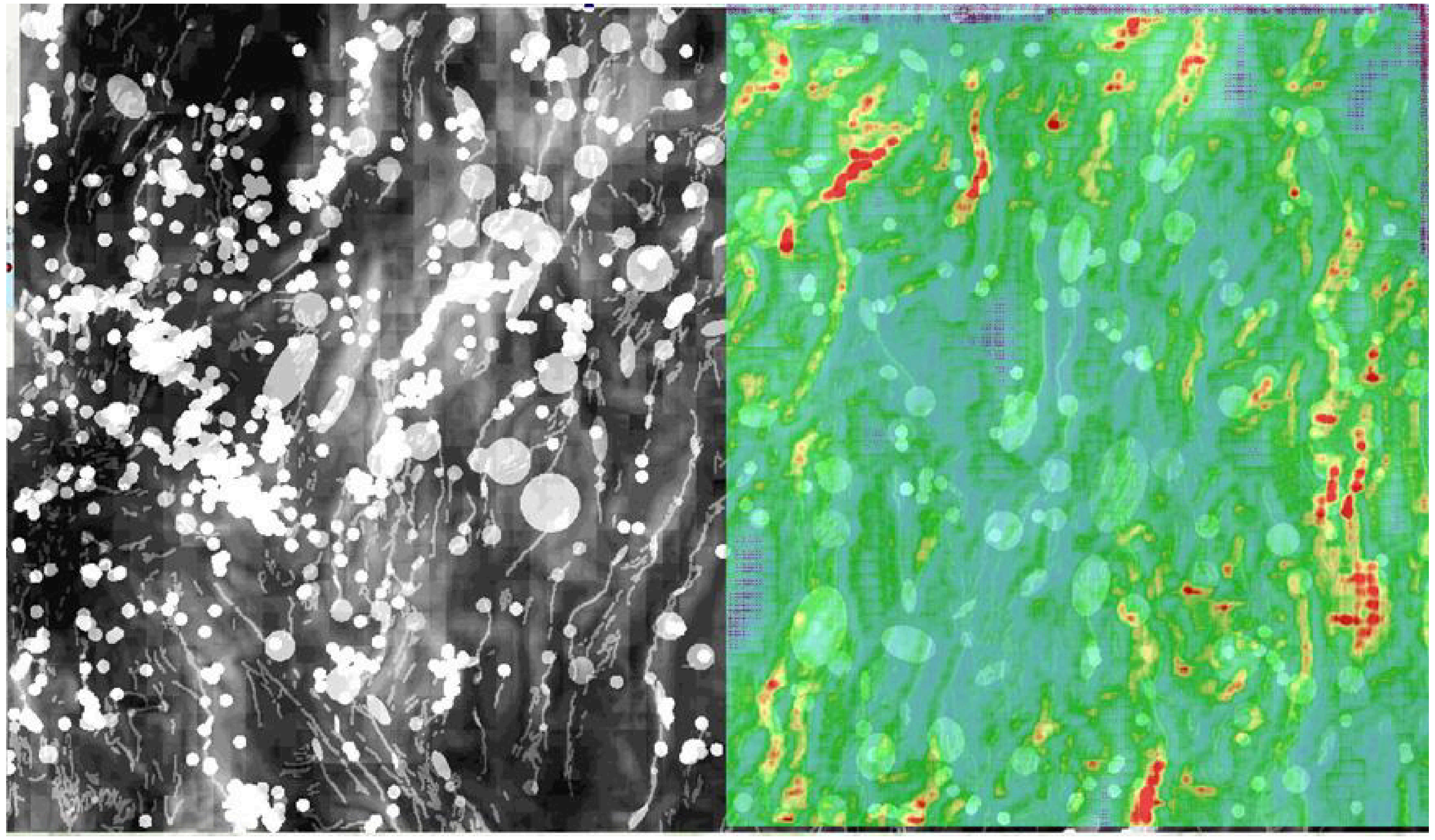
Results – Validation

Adheres to the PFA significantly

- **Learns** the PFA modeling **purely from spatial relationships** and input feature patterns.



Constructed prediction map for the right half of the map



Area of interest with Overall Favorability from PFA, given as binary layer. On top of it, the prediction of our CNN for the right half is given with transparent color mapping to compare it with the study

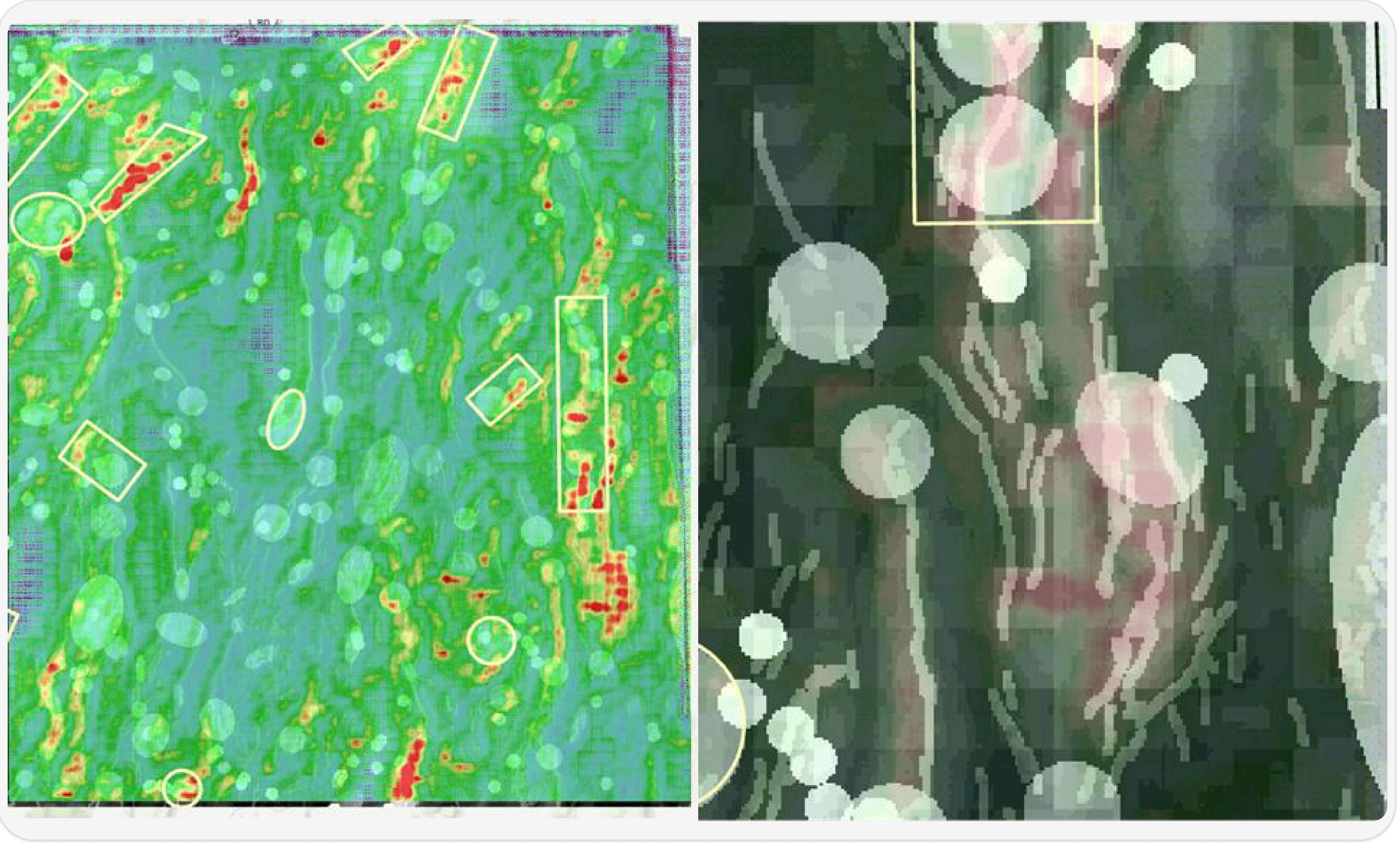
Insights

Extra confidence

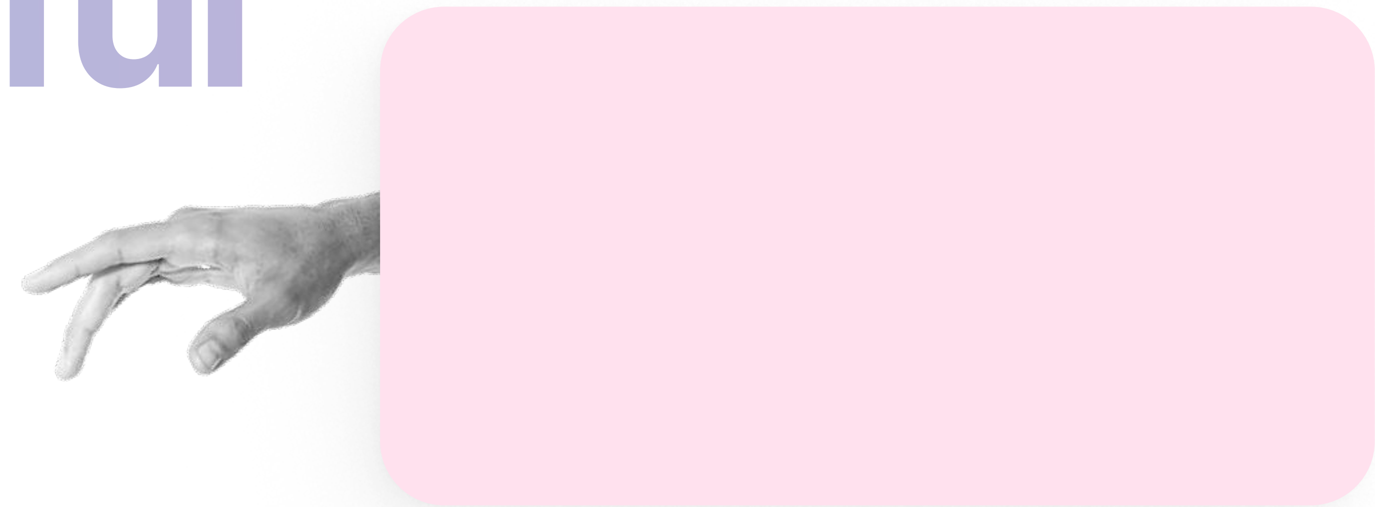
•Some of the existing fields are also identified by our model

New Insights

•**Southern tips** of the faults in the SE region of the study area are highly favored by our model



Discussion – What could be useful



Resampling strategy

- Getting rid of non-uniformity was a challenge in the short timescale of the hackathon.
- It may or may not provide better results, but it must be tried.

Feature Engineering

- Fault Density instead of Euclidean Distance.
- Alternative response features.
- Alternative predictor features.
    - DEM

Hackathon Constraints

- Many ideas, including the ones above, are left out due to time constraints.

Conclusions

- **CNN has proven to be working**
  - •Able to **capture** the **spatial patterns** and relationships that **indicate favorability**.
  - •Validates several areas from PFA.
  - •Even **suggests new areas**.

- **Fully-working pipeline**
  - •Anyone can quickly adapt this work by modifying the input-output features and testing it within seconds.

```python
# --- Seeding for reproducibility ---
SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
if torch.cuda.is_available():
    torch.cuda.manual_seed_all(SEED)
    # For full determinism (can slow down training)
    # torch.backends.cudnn.deterministic = True
    # torch.backends.cudnn.benchmark = False
print(f"Seeds set to {SEED}")

# --- Configuration ---
RASTER_DIR = "exported_layers"
RESPONSE_VAR_NAME = "smoothed_overall_favorability.tif"
INPUT_CHANNEL_NAMES = [
    "earthquake_density.tif",
    "geodetic_strain.tif",
    "horizontal_gravity.tif",
    "regional_permeability.tif",
    "temperature_at_3k.tif",
    "euclidean_distance_to_faults.tif",
]
PATCH_SIZE = 32 # ***** IMPORTANT: Ensure this matches your actual patch extraction logic if different from debug log *****
NODATA_VALUE = -9999.0

# For Resampling Strategy (Cell 5 - was Step 3 in previous numbering)
# These are for resampling based on AVG patch favorability
MANUAL_BIN_EDGES_FOR_SAMPLING = None # Will be defined in Cell 5 after inspecting data
MIN_SAMPLES_PER_RESAMPLED_BIN = 150 # Tune: Min samples to aim for in less frequent bins via oversampling
MAX_SAMPLES_PER_RESAMPLED_BIN = 600 # Tune: Max samples from any single original bin to cap dominant bins

TEST_SPLIT_FRACTION = 0.2
VALIDATION_SPLIT_FRACTION = 0.2 # of the (resampled) training data

# CNN Training Hyperparameters
LEARNING_RATE = 0.0005
BATCH_SIZE = 32
EPOCHS = 75 # Adjusted for potentially faster iteration during hackathon
PATIENCE_LIMIT = 15
CHECKPOINT_PATH_PYTORCH = 'best_pixelwise_regr_cnn_pytorch.pth'
```

.venv	6/2/2025 7:21 PM	File folder	
ArcGIS_files	5/31/2025 11:16 PM	File folder	
CNN	5/31/2025 10:20 PM	File folder	
exported_layers	6/2/2025 7:51 PM	File folder	
best_pixelwise_regr_cnn_pytorch.pth	6/2/2025 7:50 PM	PTH File	728 KB
cnn_full_pixelwise_regression_model_pyt...	6/2/2025 7:51 PM	PT File	734 KB
regression_hybrid_revised	6/2/2025 8:04 PM	Jupyter Source File	1,130 KB

A code snippet from our work (up) and the delivered folder structure of our work. We deliver all the rasters we preprocessed, the complete code for CNN, best model as a saved model file, and the generated layer as GeoTIFF

Bonus – Paper Opportunity

We consider compiling our findings into a paper

- We would love to collaborate with the geoscientists on this exciting area

A Convolutional Neural Network Approach for Enhancing Geothermal Play Fairway Analysis: A Case Study on the Nevada PFA Dataset

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Abstract

The exploration for blind geothermal systems presents significant challenges, requiring the integration of diverse geological, geochemical, and geophysical datasets. The Nevada Play Fairway Analysis (PFA) project established a comprehensive framework for this, culminating in regional geothermal potential maps. This study explores the application of deep learning, specifically Convolutional Neural Networks (CNNs), to model and potentially augment the PFA results. We utilize key predictor features from the PFA, including earthquake density, geodetic strain, gravity data, regional permeability, temperature at 3km depth, and a novel engineered feature representing Euclidean distance to faults. The target variable is a smoothed version of the PFA's overall favorability map. Our CNN, trained on the western half of the PFA study area, demonstrates a strong ability to learn the complex spatial relationships inherent in the PFA. When applied to the eastern (unseen) half, the CNN's predictions show remarkable concordance with the original PFA map, while also highlighting subtle differences and potentially new areas of interest. This research underscores the potential of CNNs as a complementary tool in geothermal exploration, capable of validating existing analyses and offering new perspectives for identifying prospective zones.

1

Thank You!

References

Faulds, J., Hinz, N., dePolo, C., Hammond, W. el al. (2015).

Nevada Great Basin Play Fairway Analysis - Reports & Appendices . [Data set]. Geothermal Data Repository. Nevada Bureau of Mines and Geology.
https://gdr.openei.org/submissions/756

Smith, C. M., Faulds, J. E., Brown, S., Coolbaugh, M. el al. (2023).

Exploratory analysis of machine learning techniques in the Nevada geothermal play fairway analysis. Geothermics, 111, 102693